

CNS Infections: Viruses I and II

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Learning Objectives

- Describe the different types of viral CNS infections and their etiology.
- Compare and contrast bacterial and viral meningitis.
- Compare the properties of the most common viral etiological agents of meningitis.
- Identify the importance of diagnostic CSF analysis.
- Illustrate the role epidemiology plays in determination of viral etiology.

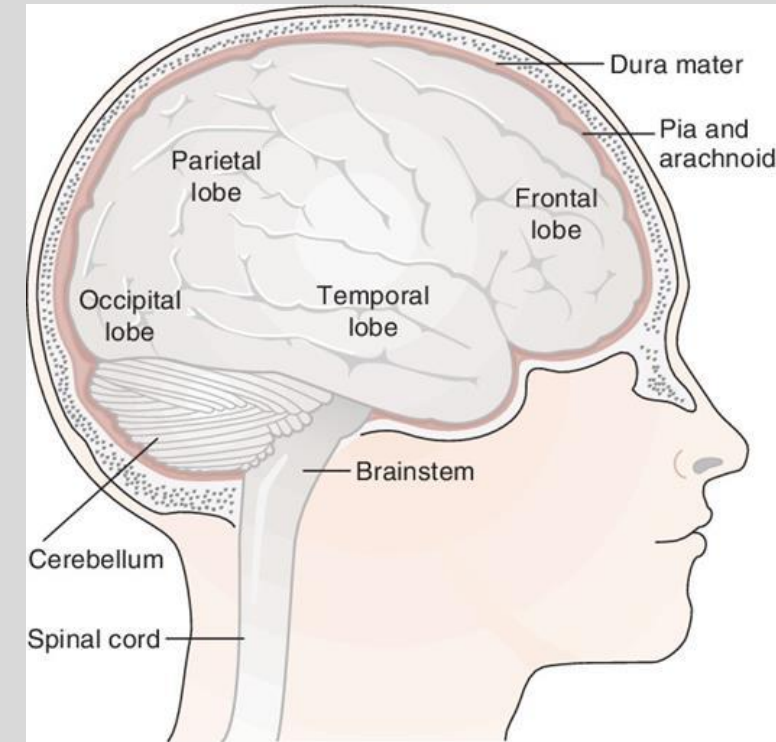
Outline

- Inflammation of the CNS (meningitis, encephalitis, myelitis)
- How CNS infections occurs
- How tissue is damaged in viral CNS infections
- Bacterial vs. Viral meningitis
- Acute viral meningitis, epidemiology, and etiologies
- Viral encephalitis, epidemiology and etiologies

Inflammation of the CNS

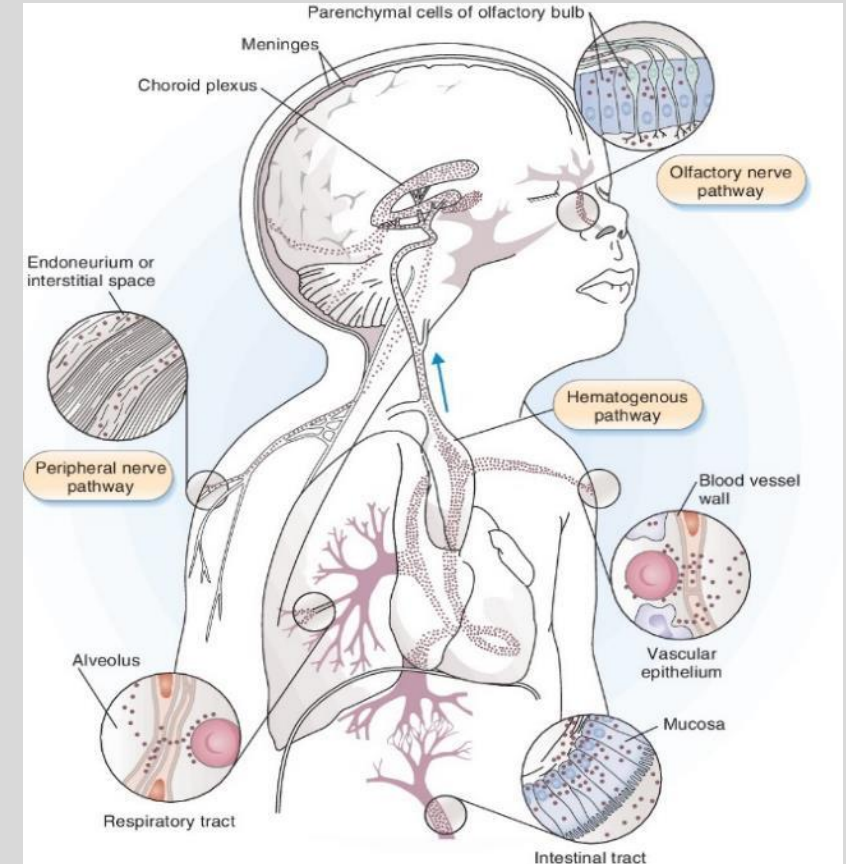
(review)

- Meningitis (meninges)
 - Acute or chronic
 - Bacterial, viral, fungal, or aseptic
 - Encephalitis (parenchyma)
 - Brain abscesses
 - Myelitis (spinal cord)
-
- *Viruses mainly cause acute meningitis & encephalitis*



How CNS Infection Occurs

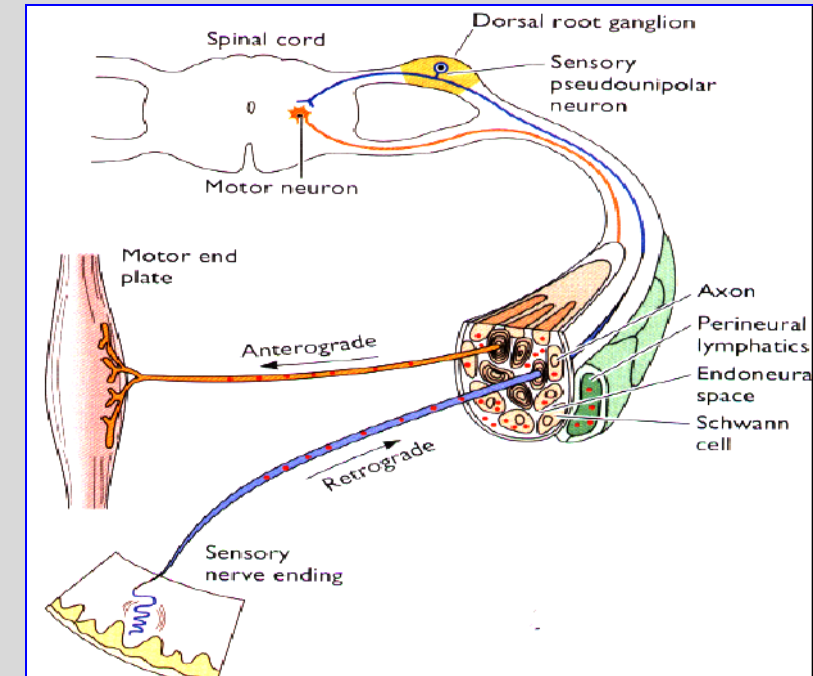
- Entry to the CNS
 - Hematogenous from inoculation at another site
 - Respiratory epithelium (aerosol, droplets; e.g. meningococci)
 - Animal bite (rabies virus), or bug bites (West Nile virus & other arboviruses)
 - Transplacentally (cytomegalovirus, rubella virus or Zika virus).
 - **from blood to CNS: choroid plexus or infection of CNS endothelial cells
- Direct inoculation
 - Trauma
 - Surgery



How Virions reach the CNS

- Neurotropism of virions

- Poliovirus for motor neurons of the spinal cord and medulla
- Mumps virus for ependymal cells lining the ventricles in the fetal brain
- Rabies virus enters the distal terminals of peripheral nerves and travels inside nerve axons via axoplasmic transport to reach nerve cell bodies in sensory ganglia and the spinal cord.
- Some viruses and other infectious agents can penetrate the body through olfactory nerve endings (direct contact with the external environment)
 - a) Nasal infections occur with rabies virus and some arboviruses under conditions with high airborne concentrations of virus, as in caves filled with infected bats.



How Virions reach the CNS

- Neurotropism of virions
 - Following chickenpox, which results from primary infection with varicella-zoster virus (VZV), virus spreads from the skin through sensory nerves to become latent in neurons in sensory ganglia. Reactivation results in spread of virus through sensory nerves to produce zoster (“shingles”)
 - HSV spreads through nerves from sites of primary infection such as oropharyngeal (primarily HSV-1) or genital mucosa (primarily HSV-2) to infect trigeminal or sacral ganglia
 - a) Virus becomes latent in sensory neurons.
 - b) Reactivation results in spread of virus along sensory nerves to mucosal surfaces, producing recurrent herpes labialis (cold sores) or herpes genitalis.

Epidemiology

- Seasonal: e.g. arboviruses spread by vectors (mosquito and ticks) more commonly in summer
- Enterovirus infections usually peak in the late summer and fall.
- Other pathogens don't have specific seasonal variation and occur with equal frequency throughout the year; e.g. HSV (most common cause of sporadic viral encephalitis in adults)

Epidemiology Changing

- Names of some arboviruses refers to the initial area of isolation (e.g., California encephalitis virus, St. Louis encephalitis virus, West Nile virus, Zika virus)
- Modern distribution much larger
- For example, although Zika virus (named for a forest in Uganda), it spread to South and Central America and throughout the Caribbean
- West Nile virus, 1st isolated from the West Nile Valley in Uganda, it is now widely distributed throughout North America and many other regions of the world.

Epidemiology

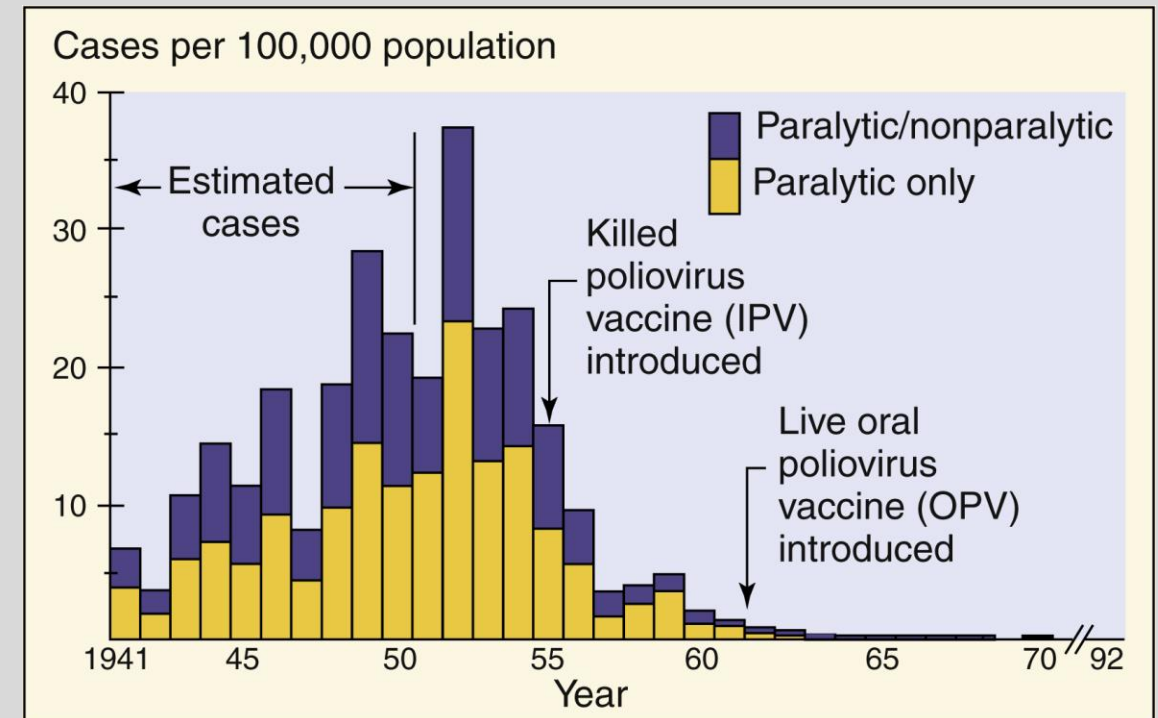
Virus	Virus Type	Geographic Location	Major Age Group	Predominant Season	Notable Features
Herpes simplex viruses 1 and 2	DNA	All	All	None	Focal symptoms and signs
West Nile virus	RNA	All	Older adults	Summer through fall	Weakness, movement disorders, poliomyelitis syndrome
St. Louis encephalitis virus	RNA	All	Older adults	Summer through fall	Most cases mild
Eastern equine encephalitis virus	RNA	Atlantic and Gulf coasts, Great Lakes	Children	Summer through fall	High mortality (50%-80%) and morbidity (80% with neurologic sequelae)
Zika virus	RNA	South and Central America, Africa, Asia	Congenital Zika syndrome. Guillain-Barré in adults	Summer through fall	Microcephaly and related defects in congenital
California (La Crosse) virus	RNA	Midwest and Northeast United States, Southern Canada	Older children	Summer through fall	<1% mortality and rare encephalitis morbidity
Enteroviruses	RNA	All	Infants and children more than adults	Summer	A71 and D68 associated with acute flaccid myelitis
Rabies virus	RNA	All	All	All	Animal bite (bats)
Varicella-zoster virus	DNA	All	Children and immunocompromised adults	Winter	
HIV	RNA	All		All	Dementia

Common Viral CNS Infection Etiologies

- Enteroviruses:
 - Coxsackievirus A and B
 - Echovirus & Enterovirus
- Arboviruses:
 - Eastern and Western equine encephalitis virus
 - LaCrosse encephalitis virus
 - St. Louis encephalitis virus
- Herpes Simplex Virus (HSV)
- Varicella-zoster virus (VZV)

Less Common Viral CNS Infection Etiologies

- JC Virus: it causes Progressive Multifocal Leukoencephalopathy (PML)
- Rabies virus
- Lymphocytic choriomeningitis virus (LCMV)
- Polio virus: a previous cause of infantile paralysis; preventable with a vaccine
- sIgA is potent aspect of the IR



Acute Viral Meningitis

- Viruses leading cause of aseptic meningitis
- Symptoms:
 - fever,
 - rash,
 - headache,
 - nuchal rigidity,
 - and photophobia, often with
 - an increased number of lymphocytes in CSF (lymphocytic pleocytosis)

Bacterial vs. Viral Meningitis

Acute Meningitis

	Bacterial	Viral
Presentation	Focal or global	Mostly focal
CSF	↑ Opening pressure ↑ PMNs ↑ Protein ↓ Glucose	Normal/↑ Opening pressure ↑ Lymphocytes Normal protein Normal/↑ glucose
Outcomes	High mortality and morbidity	Low mortality and morbidity
Tx	Immediate abx therapy	Supportive therapy

*Note: In infants empirical antimicrobial is initiated even if viral is suspected

Bacterial vs. Viral Meningitis

CSF Composition with numerical measurements

Usual Composition of Cerebrospinal Fluid in Various Central Nervous System Infections

Factor	Normal	Acute Bacterial Meningitis	Chronic Mycobacterial and Fungal Meningitis	Acute Viral Meningitis	Viral Encephalitis	Brain Abscess
Leukocytes (per mm ³)	0–6	>1,000	100–500	10–1,000, usually <300	10–500	10–100
% Neutrophils	0	>50	<10	May predominate in first 24 h, then <50	May predominate in first 24 h, then <50	<50
Red blood cells (per mm ³)	0–2	0–10	0–2	0–2	10–500 (HSV)	0–2
Glucose (mg/dL) ^a	40–80	<30	≤40	40–80	40–80	40–80
Protein (mg/dL)	20–50	>100	50–100 (fungal), >100 (mycobacterial)	50–100	50–100	50–100

^aDiagnostic values are best interpreted as the ratio of glucose levels in CSF compared with those in serum. Ratios <0.4 are considered diminished.

Diagnosis

- **CSF**: The typical profile is a **lymphocytic pleocytosis** (25 to 500 cells/ μ L), a normal or slightly elevated protein concentration [50-100 mg/dL], a normal glucose concentration, and a normal or mildly elevated opening pressure. A matching CBC will often parallel the CSF findings ($\uparrow$ Lymphocytes & Monocytes)
- No organisms on Gram's or Auramine-Rhodamine stained smears or India ink preparations of CSF
- Cultures of the CSF for bacteria or fungi are negative
- Nucleic Acid Amplification (NAA): e.g. Enterovirus Consensus RT-PCR kit



<https://www.elitechgroup.com/molecular-diagnostics-intl/product/enterovirus-elite-mgb-kit/>

Review Question 1

Choose from the options in parentheses to complete the sentences or fill in the blanks:

In a suspected case of meningitis, CSF containing high protein and low glucose may indicate (**bacterial/viral**) acute meningitis which requires _____ treatment.

In a suspected case of meningitis, CSF containing high lymphocytes and normal glucose may indicate (**bacterial/viral**) acute meningitis which requires _____ treatment.

Viral Encephalitis

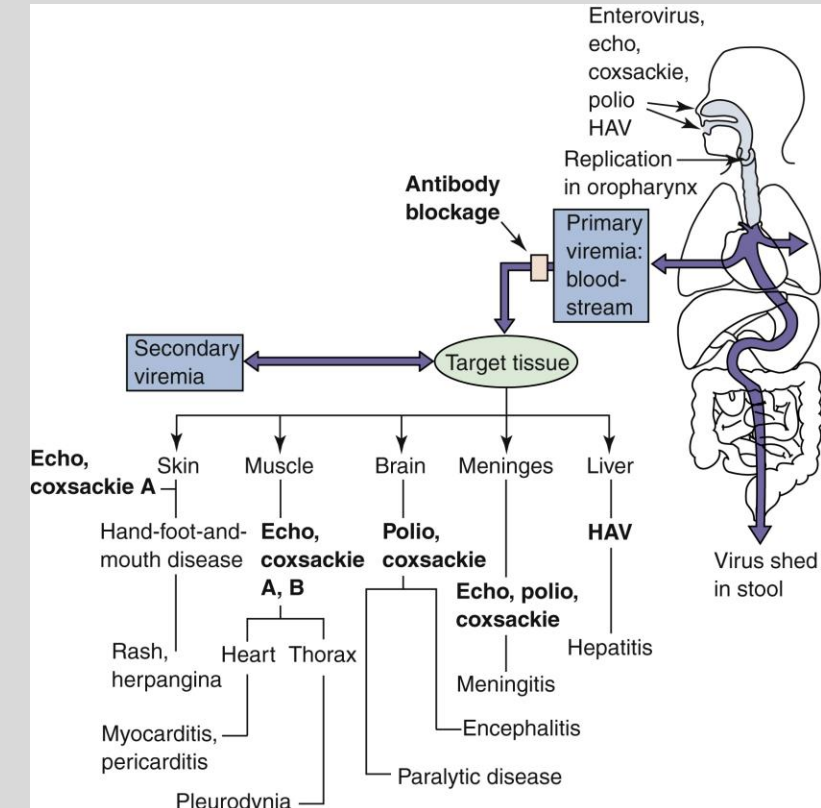
- Encephalitis is usually caused by viruses
- Brain parenchyma
- Encephalitis may have associated meningitis (meningoencephalitis) and, in some cases, involvement of the spinal cord or nerve roots (encephalomyelitis, encephalomyeloradiculitis)
- Sporadic: HSV, VZV and, less commonly, enteroviruses.
- Epidemics of encephalitis are caused by arboviruses, which belong to several different viral taxonomic groups including Alphaviruses (e.g. Eastern equine encephalitis virus, Western equine encephalitis virus), Flaviviruses (e.g., WNV, St. Louis encephalitis virus, and Bunyaviruses (e.g., California encephalitis virus serogroup, LaCrosse virus).

Viral Encephalitis

- CSF analysis very similar to acute meningitis
- Confirmation by NAA

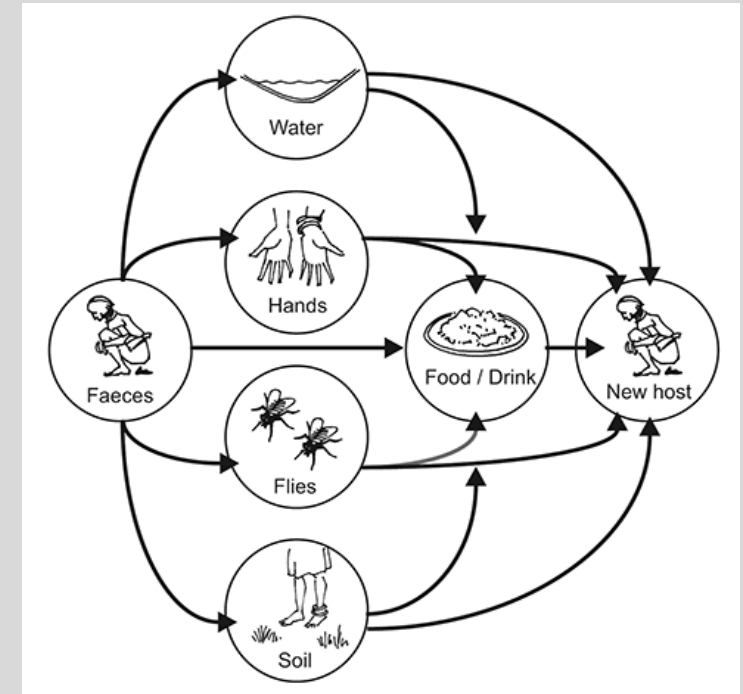
Coxsackievirus A & B, Echovirus, & Enteroviruses

- Enteroviruses are small (picornaviridae), non-enveloped, +ssRNA viruses with an icosahedral capsid; lytic
- Most common cause of acute viral meningitis
- Enteroviruses enter & multiply in the GI track (not a prominent cause of gastroenteritis)
- Multiple serotypes:
 - 3 serotypes of poliovirus,
 - 24 serotypes of coxsackievirus A,
 - 6 serotypes of coxsackievirus B,
 - 34 serotypes of echovirus, and
 - Enterovirus 4.



Enteroviruses Transmission

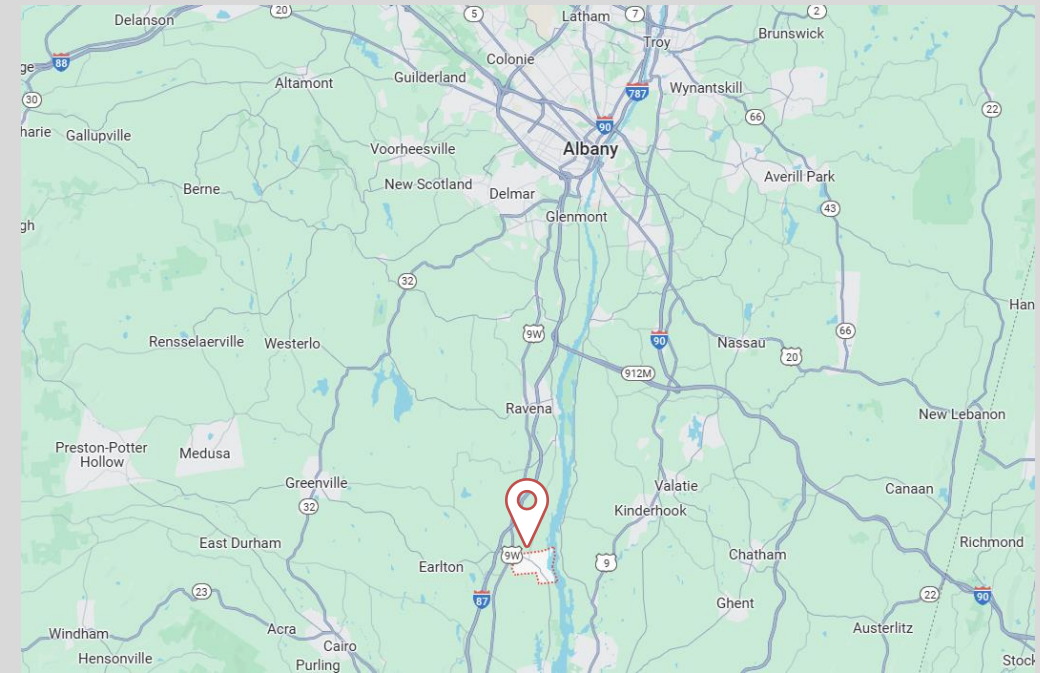
- Transmission by fecal-oral route (food, water, hands), enter the body through the G.I. tract, spread to the blood stream (producing viremia), and infect other organs including the meninges
- The viruses are stable and can survive in chlorinated swimming pools, the GI tract (stomach acid, proteases and bile), and sewage systems



<https://epidemics.ifrc.org/volunteer/training-manual/group-1-faecal-oral-infections>

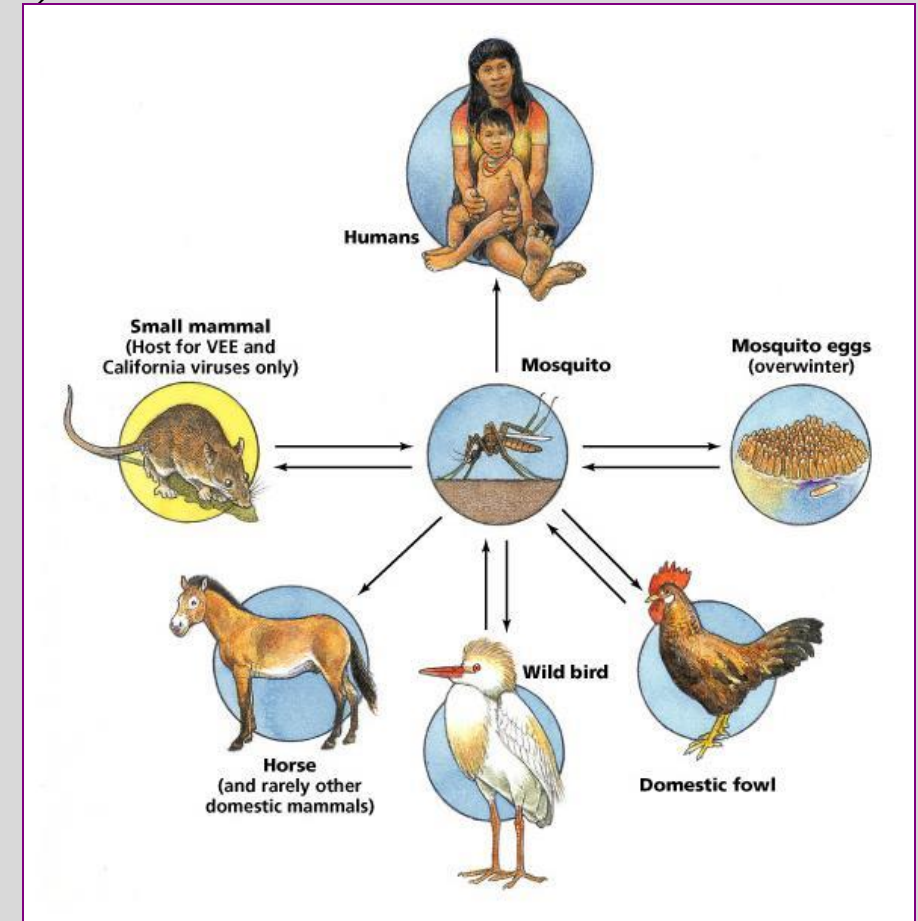
Coxsackieviruses

- Etiologic agent of foot and mouth disease & Herpangina (vesicular diseases)
- Can cause acute viral meningitis
- Diagnosis by NAA
- Disease typically self limiting unless the illness is associated with encephalitis (meningoencephalitis) or occurs in children younger than 1 year



Arboviruses

- Vector-borne illnesses (mostly mosquitoes)
- Examples:
 1. Alphaviruses
 1. Eastern Equine encephalitis virus
 2. Western Equine encephalitis virus
 2. Flaviviruses
 1. St. Louis encephalitis virus
 2. West Nile virus
 3. Japanese encephalitis virus
 3. Bunyaviruses
 1. LaCrosse encephalitis virus
- Most individuals are asymptomatic
- Few develop serious life threatening illness



Arboviruses (EEE)

- Eastern equine encephalitis (EEE) is caused by a virus transmitted to humans and equines (horses) by the bite of an infected mosquito.
- Incubation period: 4-10 days
- Symptoms: sudden onset of fever, myalgia, and a headache of increasing severity
- 1/3 with clinical encephalitis caused by EEE will die from the disease
- Endemic in the New England states

Arboviruses (WEE)

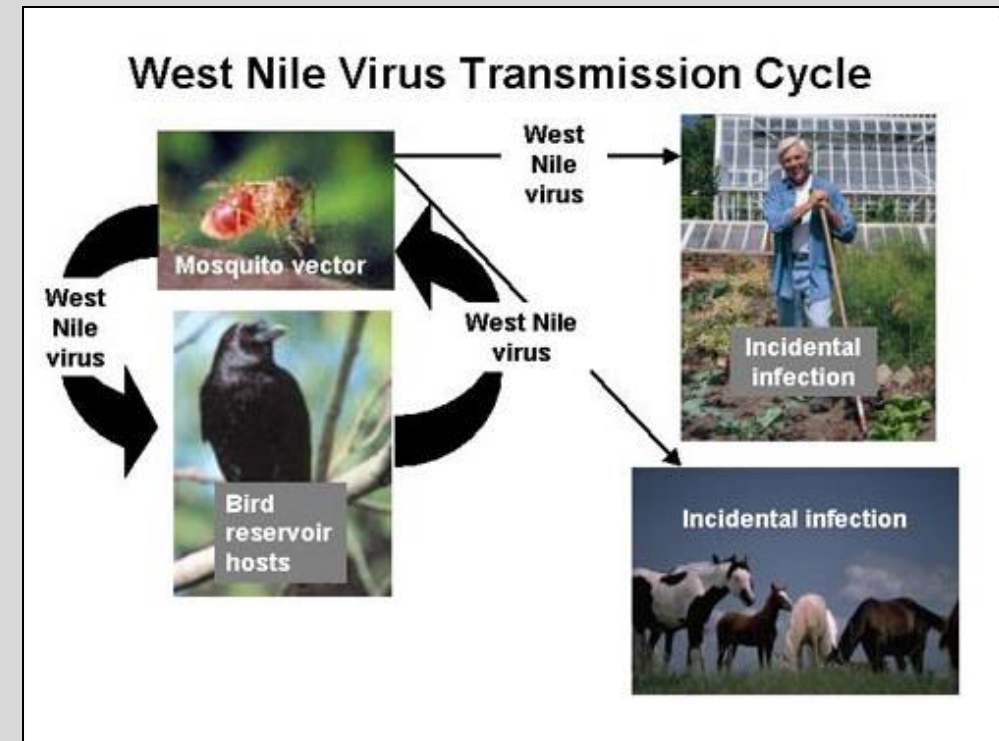
- Western equine encephalitis (WEE) virus is an important cause of encephalitis in horses and humans in North America, mainly in western parts of the USA and Canada.
- The life cycle of WEE involves birds, in which the infection is inapparent, and mosquitoes.
- Symptoms: sudden onset with fever, headache, nausea, vomiting, anorexia and malaise, followed by altered mental status, weakness and signs of meningeal irritation

Arboviruses (SLE)

- In the US, the leading cause of epidemic Flavivirus encephalitis is St. Louis encephalitis (SLE) virus
- SLE is the most common mosquito-transmitted human pathogen in the U.S
- SLE virus is distributed throughout the lower 48 states.
- Since 1964, there have been ~5000 confirmed cases of SLE with an average of 193 cases per year (range 4 - 1,967)
- Illness ranges in severity from a simple febrile headache to meningoencephalitis, with an overall case-fatality ratio of 5-15 %

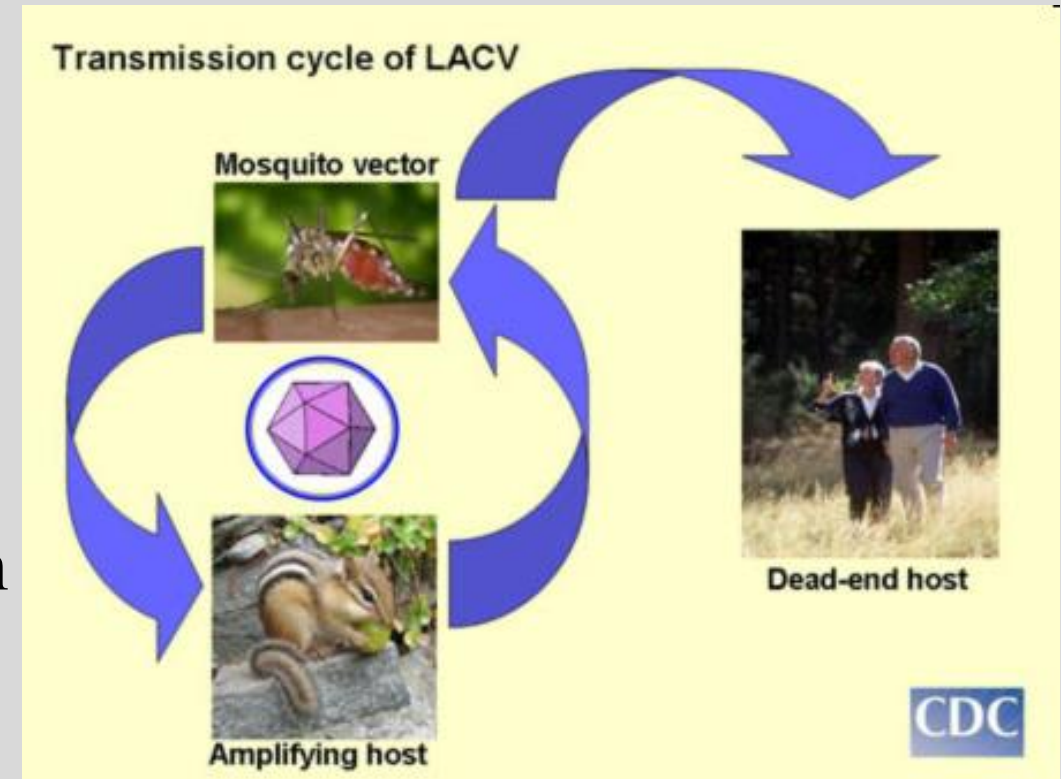
Arboviruses (WNE)

- West Nile virus was first isolated in the West Nile Province of Uganda in 1937
- The first outbreak of WNV encephalitis occurred in NY City (mostly in Queens) and in a few neighboring areas in late August and Sept. 1999.
- International travel and climate change played a role in its introduction to North America



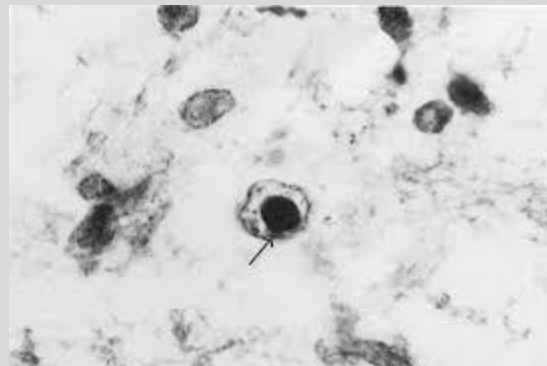
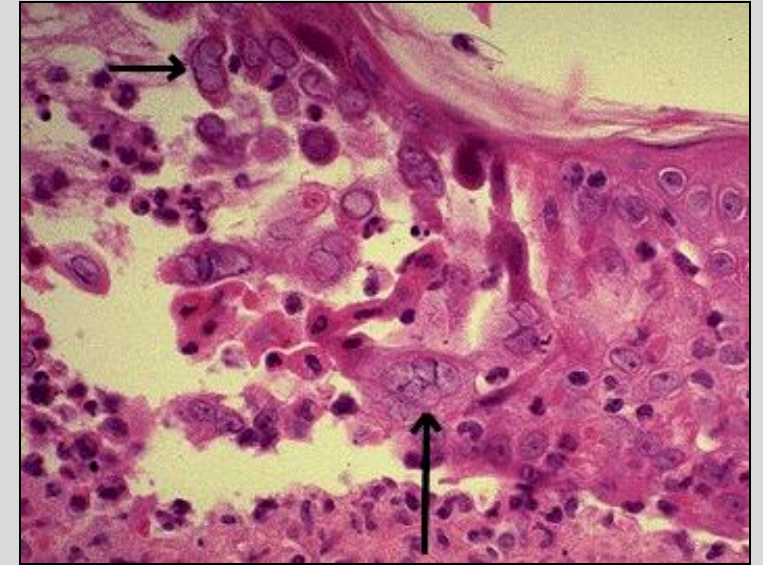
Arboviruses (LAC)

- La Crosse (LAC) encephalitis was first discovered in La Crosse, Wisconsin in 1963
- The virus has been identified in several Midwestern and Mid-Atlantic states
- About 75 cases of LAC encephalitis are reported to the CDC each year
- Most cases of LAC encephalitis occur in children under 16 years of age
- LAC virus cycles between mosquitoes and vertebrate hosts (chipmunks, tree squirrels) in forest habitats



HSV 1 and 2

- Herpes simplex viruses (HSV) 1 and 2 have similar but not identical genomes
- Enveloped ds DNA viruses with an icosahedral nucleocapsid
- Cytopathic effects: Cowdry type A intranuclear inclusion bodies & giant multinucleated cells (Tzanck smear w Giemsa stain)

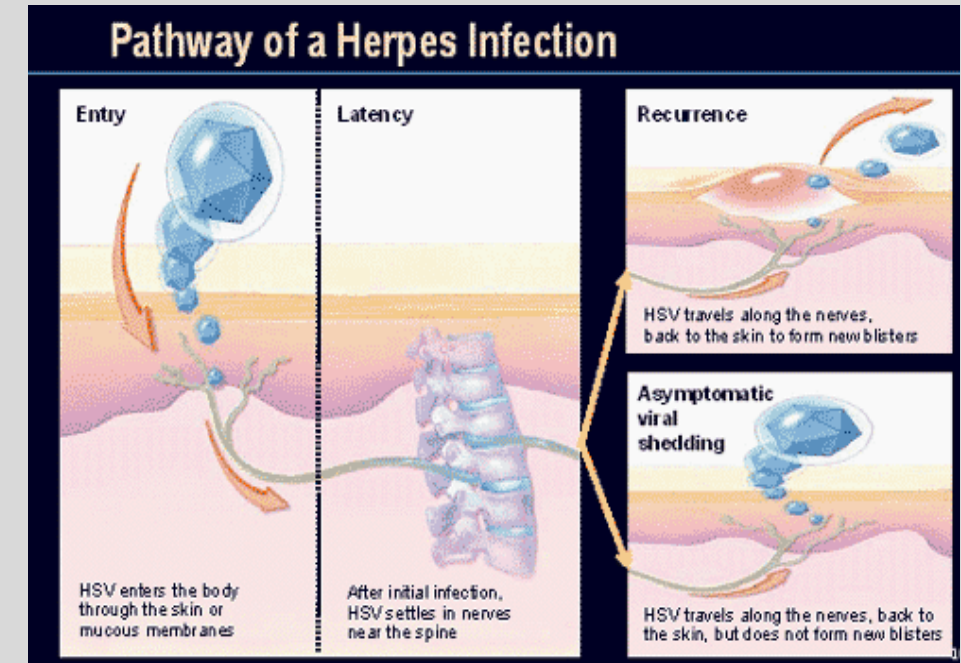


HSV 1 and 2

- Both HSV types can infect a variety of cells including epithelial cells at the site of entry on the mouth (cold sores) and genital areas (causing raised, fluid-filled vesicles)
- These cells can be visualized microscopically using a test called a Tzanck smear (stained smears of scrapings or swabs of the base of the lesion – Giemsa is the preferred stain)
- Confirmation by PCR
- Serology is not useful due to a high seroprevalence in the population

HSV 1 and 2

- Following primary infection around the mouth or genitals, HSV infects the innervating neuron, traveling by retrograde transport to the ganglion where they establish a latent infection: **the trigeminal ganglia for oral HSV, and the sacral ganglia for genital HSV**
- Both latent virions can reactivate in the ganglia, replicate and travel down the axons causing vesicular lesions containing infectious virus (cold sore or genital sores)



HSV encephalitis

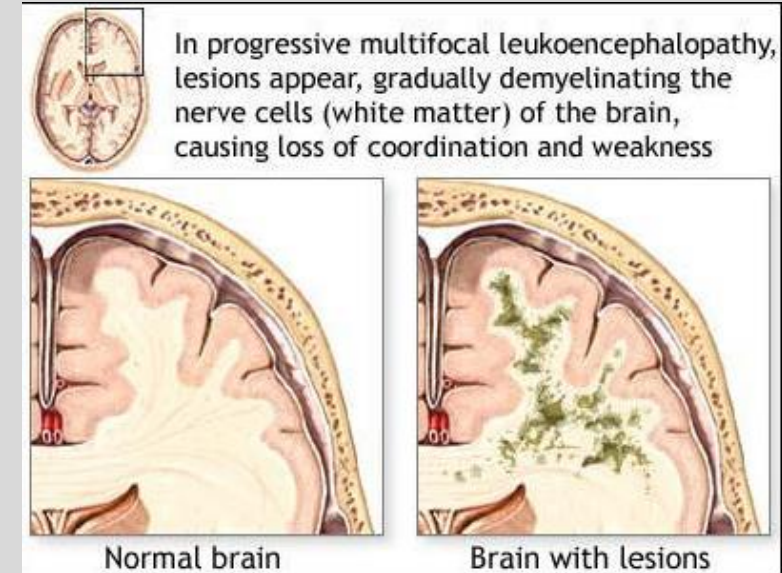
- Adult and pediatric (post-neonatal) HSV encephalitis is usually caused by HSV-1; HSV-1 is the most common viral cause of sporadic viral encephalitis
- Reactivated HSV can spread either to the skin, along the branches of the trigeminal nerve, causing sores on the lips (herpes labialis), or to the brain, infecting the meninges
- Patients often display bizarre behavior, personality changes, insomnia, and gustatory hallucinations
- 70% of neonatal HSV infections are caused by HSV-2 (herpes genitalis), which is acquired during vaginal delivery by contact with secretions in the infected birth canal
- Untreated, neonatal HSV infection can disseminate to the CNS in most cases
- Antiviral therapy: acyclovir and its derivatives, valacyclovir, penciclovir and famciclovir

Varicella-zoster virus (VZV)

- Varicella-zoster virus causes chicken pox & herpes zoster or shingles in reactivation
- Chicken pox is acquired by inhalation but also by contact with skin vesicles
- Virus replicates in the respiratory tract then disseminates via bloodstream to skin producing the characteristic fluid filled vesicles of chickenpox
- The virus then travels from the skin up sensory nerves by retrograde axonal transport and becomes latent in the nerve cell bodies of the dorsal root or cranial nerve ganglia
- Very similar to HSVs (latency, reactivation, neurotropism)
- Zostavax (>60yrs) & Shingrix vaccines (> 50 yrs)

JC virus

- JC virus is a human polyoma virus nonenveloped, ds DNA virus with an icosahedral capsid
- The JC virus enters through the respiratory tract and then spreads to the kidneys
- B cells and monocytes are infected (latency)
- Reactivation occurs in immunocompromised patients where the virus spreads through the blood to CNS (Progressive multifocal leukoencephalopathy; PML)
- PML is a subacute demyelinating & necrotic disease.
- The disease progresses to include paralysis of the arms and legs and death.

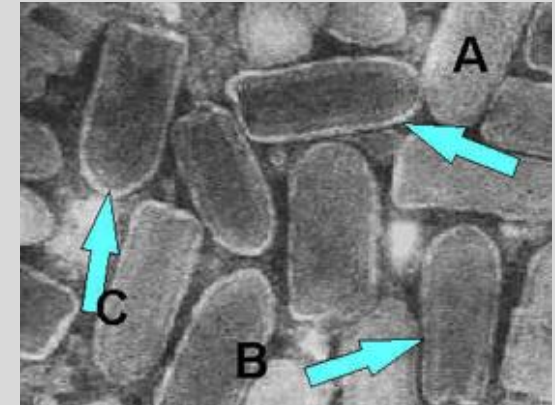


JC virus

- Approximately 10% of people with AIDS develop PML (85% of all cases) with a 90% mortality rate
- Case have decreased with the success of ART
- Diagnosis is aided by CT/MRI imaging & PCR of CSF sample

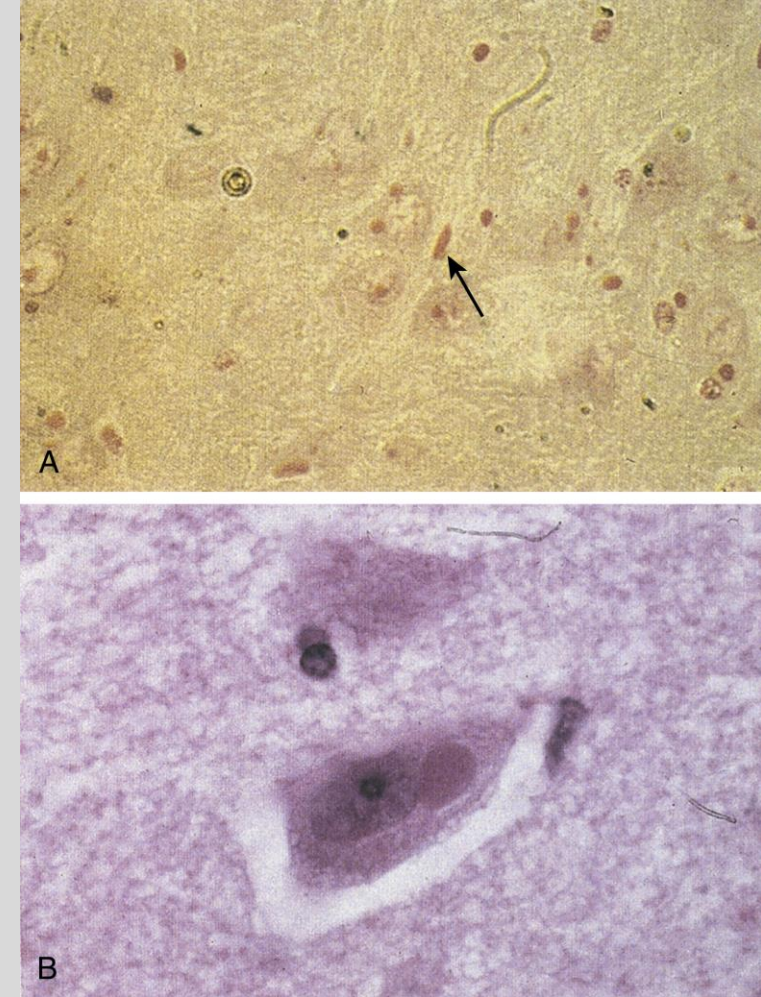
Rabies Virus

- Rhabdoviridae; Bullet-shaped, enveloped, -ve sense ssRNA virus
- Rabies virus causes an acute encephalitis in all warm-blooded hosts, including humans, and the outcome is almost always fatal (Zoonosis)
- Although all species of mammals are susceptible to rabies virus infection, only a few species (esp. racoons & bats) are important as reservoirs for the disease
- Transmission is through a bite containing virus-containing saliva of an infected animal
- After uptake into peripheral nerves, virus transported to CNS via retrograde axoplasmic flow
- During the period of cerebral infection, the classic behavioral changes associated with rabies develop



Rabies Virus

- Rabies infection frequently causes cytoplasmic eosinophilic inclusion bodies (**Negri bodies**) in neuronal cells, especially pyramidal cells of the hippocampus and Purkinje cells of the cerebellum
- Ag detection using direct immunofluorescence
- RT-PCR from multiple samples are needed to confirm the infection (serum, CSF, skin)



Negri bodies caused by rabies. (A) A section of brain from a patient with rabies shows Negri bodies (arrow) . (B) Higher magnification from another biopsy specimen.

Review Question

A 20-year-old male college student presents to the ED with nuchal rigidity, fever, and headaches for the past 3 days.

He was accompanied by his roommate who claims the patient has had multiple periods of confusion since his symptoms began. His vaccination status is unknown, and he has no significant past medical history. A culture of his CSF is taken, and antibiotics are administered.

Microscopic analysis of the CSF culture reveals gram negative diplococci.

What bacteria is the most likely etiology of this patient's meningitis?

Summary Slide

- Limited access to CNS by microbes
- CNS bacterial infections: meningitis and abscess
- Often severe to life threatening
- Immediate action (relief pressure, stabilize, draw CSF, empirical antibiotic)
- CSF microbiology (collect prior to antibiotic administration) and chemistry evaluation is crucial
- Vaccination!

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Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>